



IBPI | Yuan Ze University

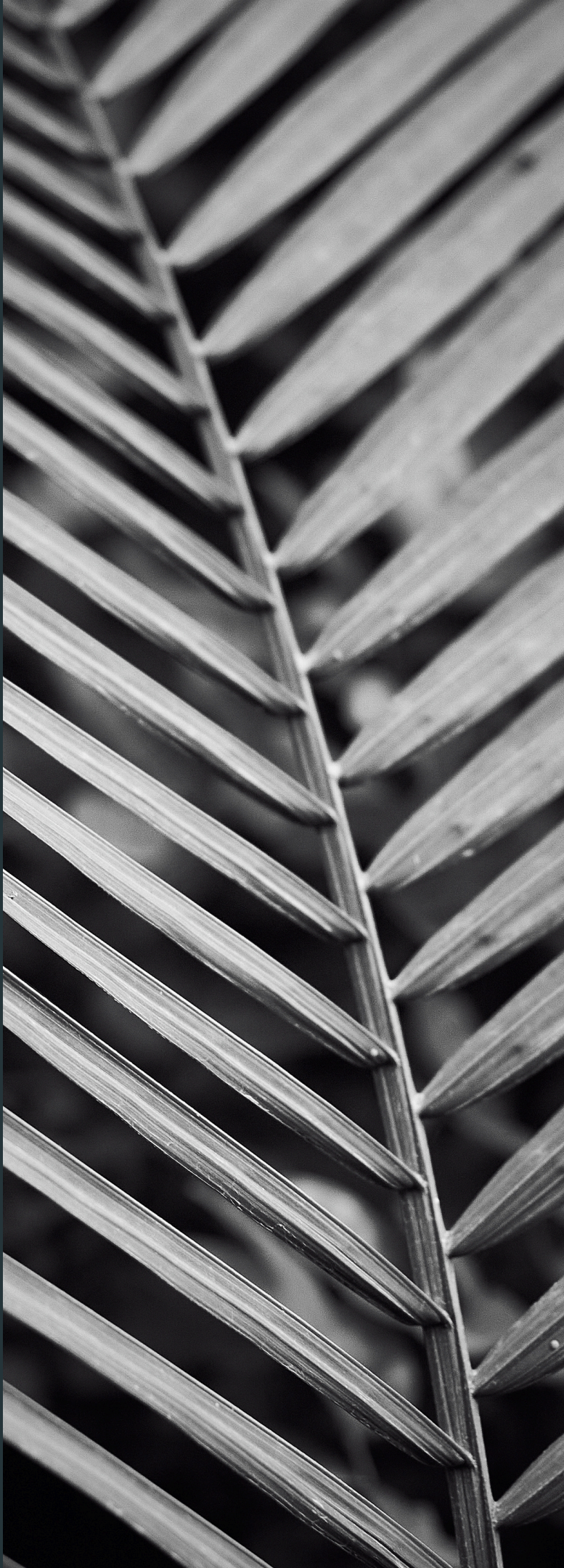


Comparative Analysis of FGSM, PGD, and Ensemble PGD on Image Classification Models

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Introduction

Deep neural networks have become one of the most successful technologies in computer vision.

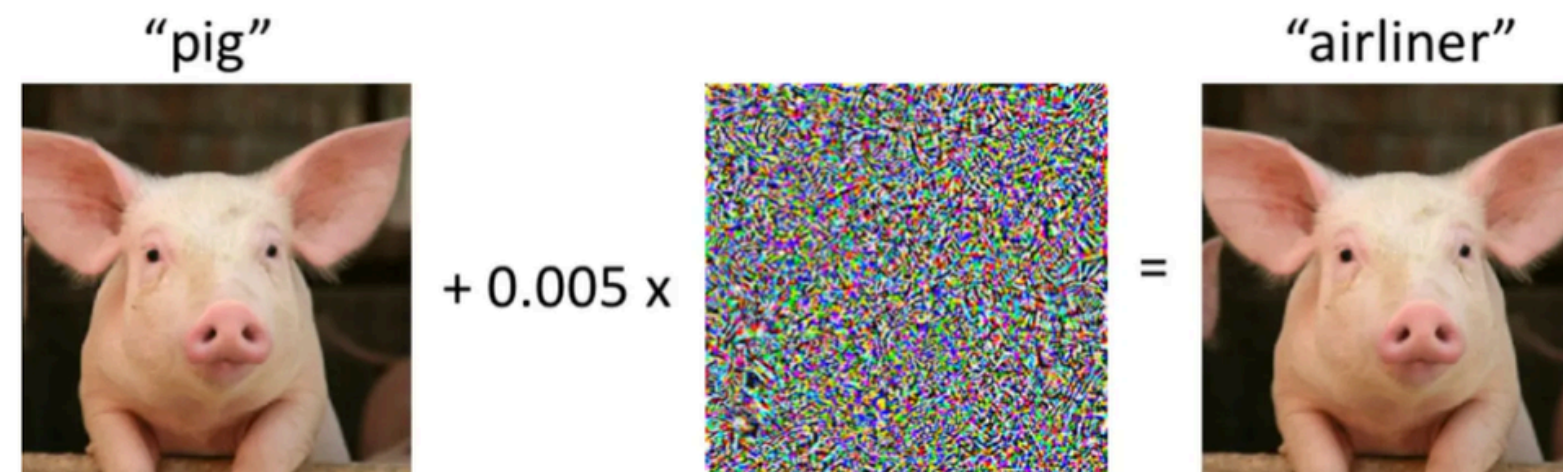
Applications include:

- Medical diagnosis
- Autonomous vehicles
- Face recognition
- Security systems
- Image search
- Industrial inspection

Problem Statement

Although neural networks achieve high accuracy, they have an important weakness.

Small, carefully designed perturbations can cause completely incorrect predictions.

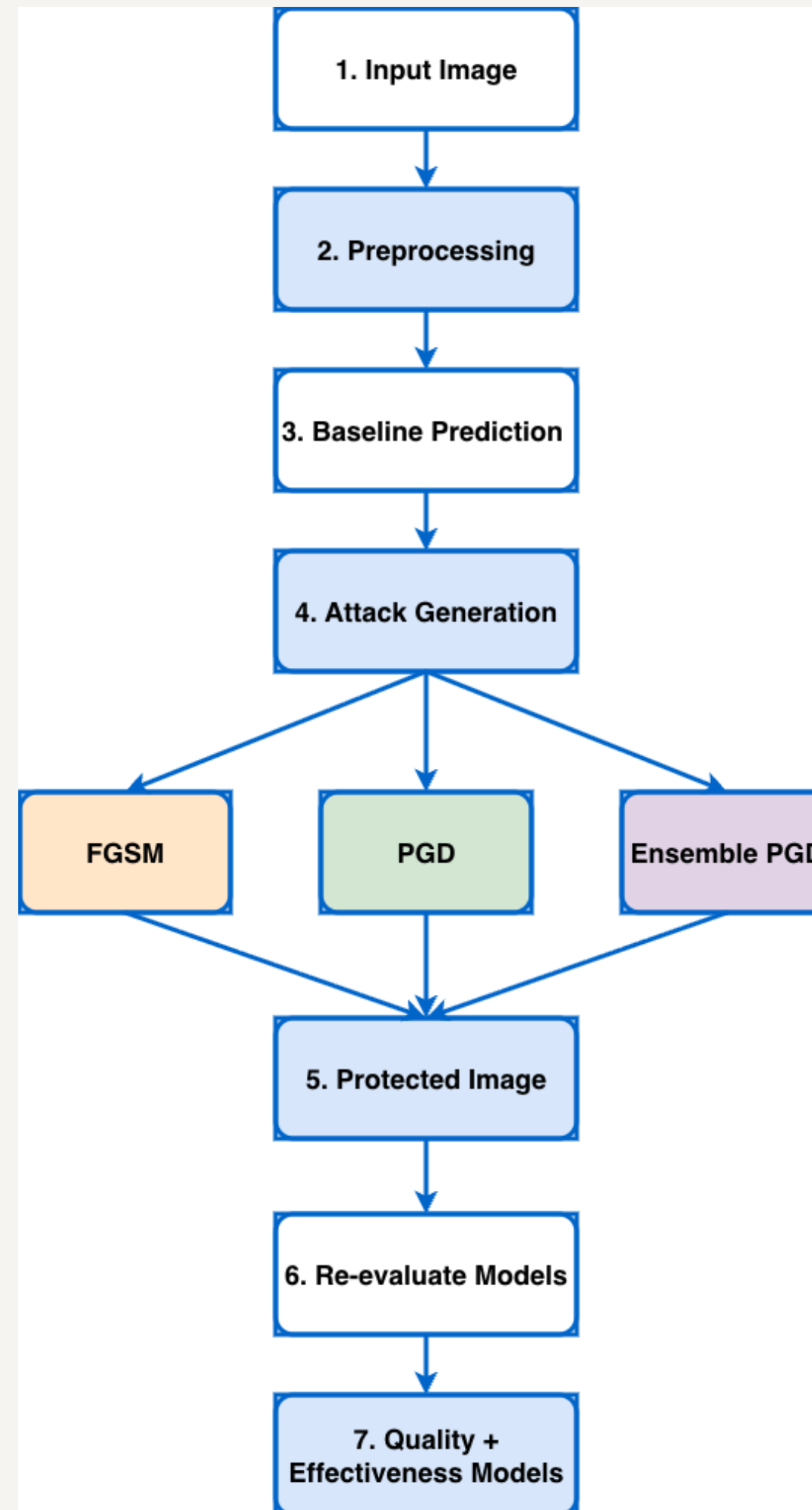


Example of misclassification due to Adversarial Attacks

Research Objectives

- ▶ Implement Fast Gradient Sign Method (FGSM)
- ▶ Implement Projected Gradient Descent (PGD)
- ▶ Develop an Ensemble PGD approach using multiple neural networks
- ▶ Compare attack effectiveness across different architectures
- ▶ Evaluate visual quality using PSNR and SSIM
- ▶ Measure adversarial transferability

Data Flow





Theoretical Framework

Fast Gradient Sign Method (FGSM) single-step

The fast gradient sign method works by using the gradients of the neural network to create an adversarial example.

$$adv_x = x + \epsilon * \text{sign}(\nabla_x J(\theta, x, y))$$

Projected Gradient Descent (PGD) iterative

PGD is an iterative, gradient-based method that projects perturbations back into a constrained set and remains a standard robustness benchmark.

$$x^{(k+1)} = P_C \left(x^{(k)} - \alpha_k \nabla f(x^{(k)}) \right)$$

Ensemble Projected Gradient Descent

Traditional PGD:

Optimize against only one model. Example: Resnet50

Proposed Ensemble PGD:

One perturbation is optimized simultaneously for three models:
ResNet50, EfficientNet-B0, and ViT-B/16

Significance of the Study

Why combine these three architectures?

ResNet50

- Residual CNN

ViT-B/16

- Transformer architecture

EfficientNet-B0

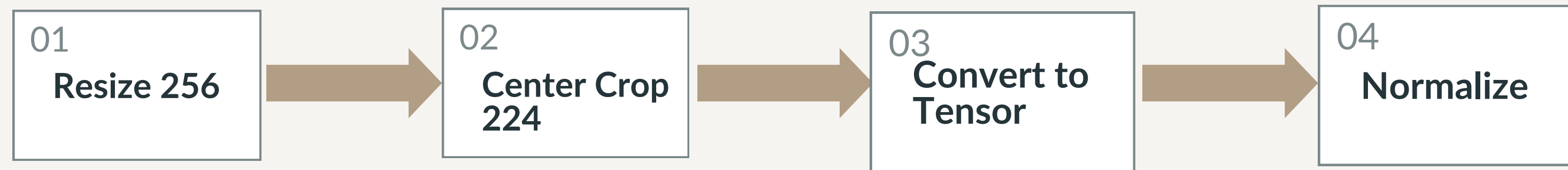
- Compound-scaled CNN

The objective is to improve adversarial transferability instead of optimizing for only one architecture.



Methodology

Preprocessing:



Attack Generation:

FGSM
PGD
Ensemble PGD

Evaluation Models:

- ResNet50
- EfficientNet-B0
- ViT-B/16

Metrics

- PSNR
- SSIM
- MSE
- L2 Norm
- Attack Success Rate
- Confidence Drop

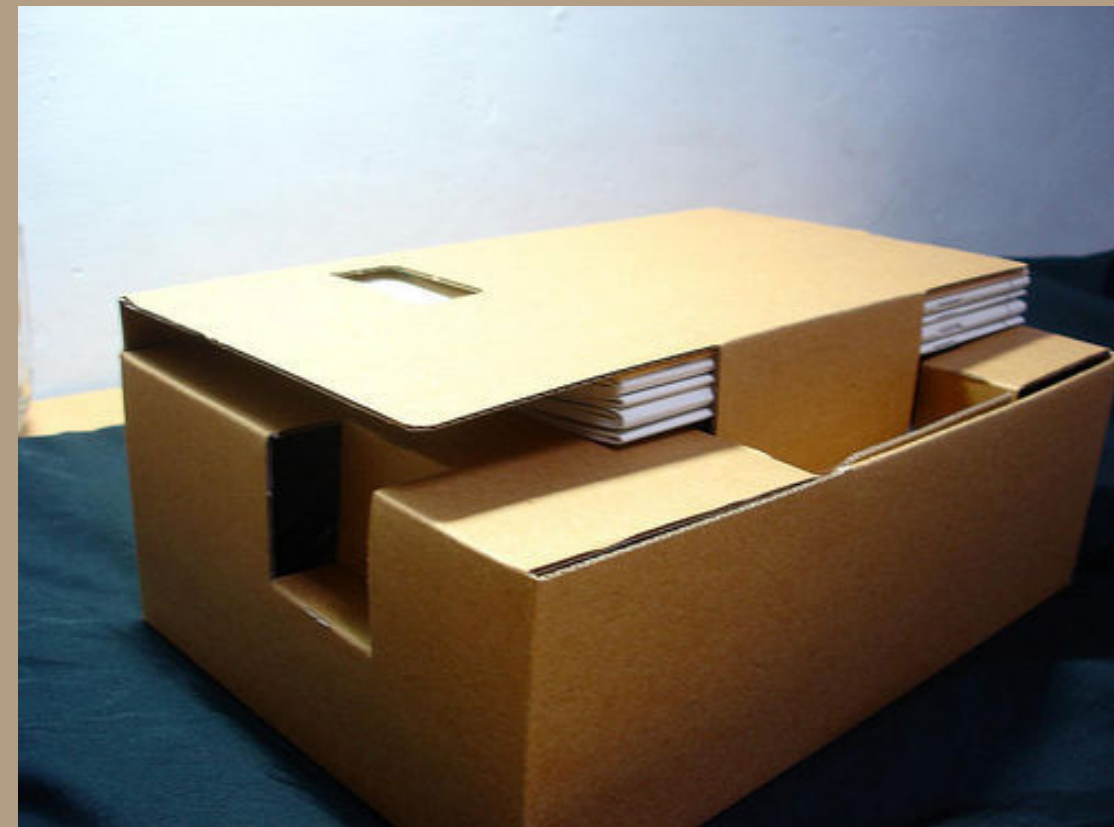
Dataset:

ILSVRC2012 Validation Images

Experimental Setup

- **Dataset:** 20/1000 ImageNet validation images
- **Epsilon values:** $4/255$, $8/255$, $16/255$
- **PGD iterations:** 20
- **Three evaluation models.**

original image from ILSVRC2012 validation dataset



Epsilon 8/255

FGSM

Original predictions before protection:

```
resnet50..... carton (49.17%)  
vit..... carton (90.10%)  
efficientnet..... carton (81.00%)
```

Model-by-model before/after view:

```
resnet50: carton (49.17%) -> carton (14.18%) [same] ✖  
vit: carton (90.10%) -> carton (89.19%) [same] ✖  
efficientnet: carton (81.00%) -> carton (90.31%) [same] ✖
```

PGD

Model-by-model before/after view:

```
resnet50: carton (49.17%) -> paper towel (1.13%) [CHANGED] ✔  
vit: carton (90.10%) -> carton (87.41%) [same] ✖  
efficientnet: carton (81.00%) -> carton (86.14%) [same] ✖
```

Ensemble PGD

Model-by-model before/after view:

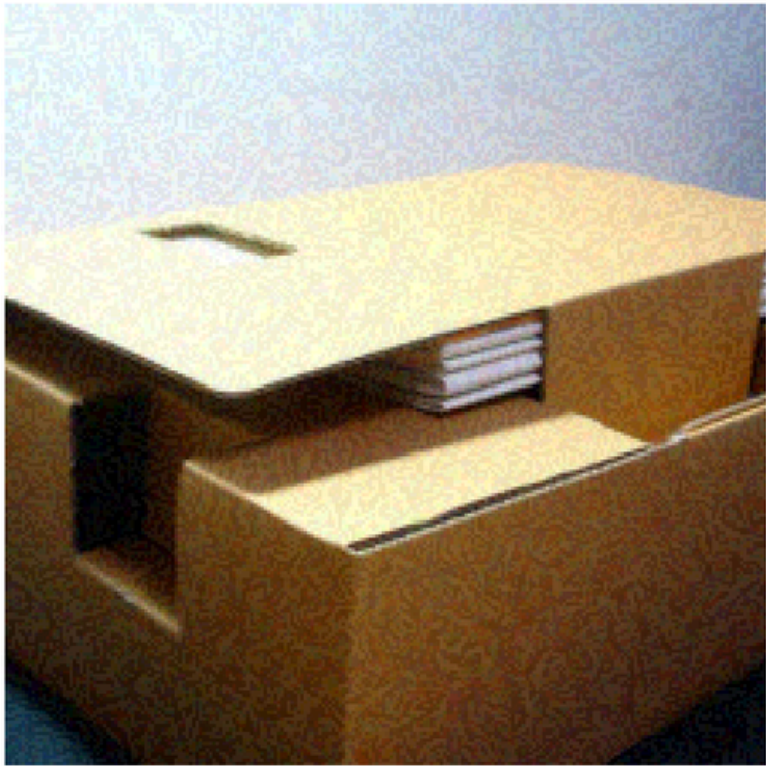
```
resnet50: carton (49.17%) -> wombat (0.43%) [CHANGED] ✔  
vit: carton (90.10%) -> desk (35.65%) [CHANGED] ✔  
efficientnet: carton (81.00%) -> desk (1.60%) [CHANGED] ✔
```

Attack Method Comparison on Cropped 224x224 Image ($\epsilon = 8/255$)

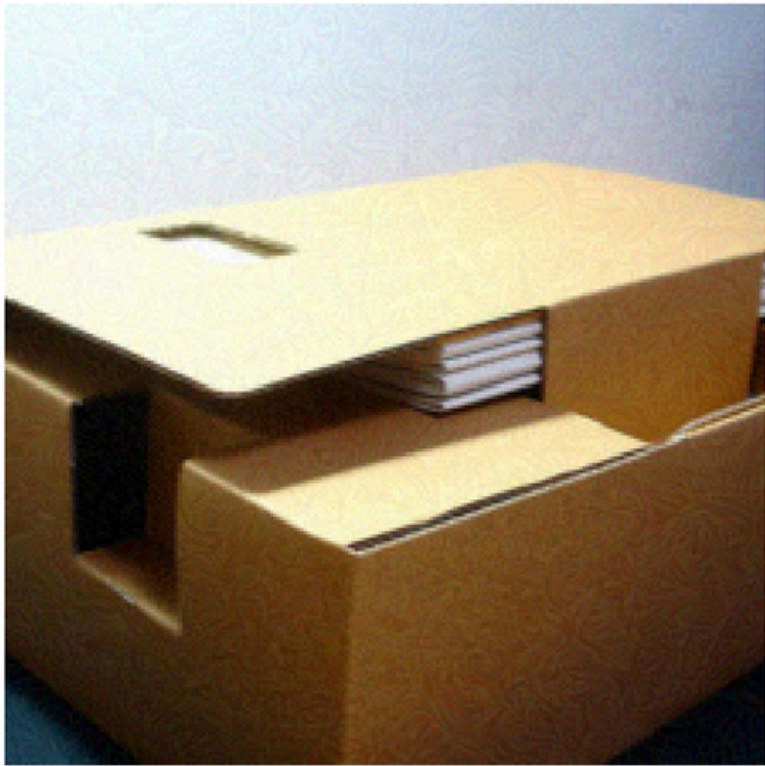
Original Cropped 224x224



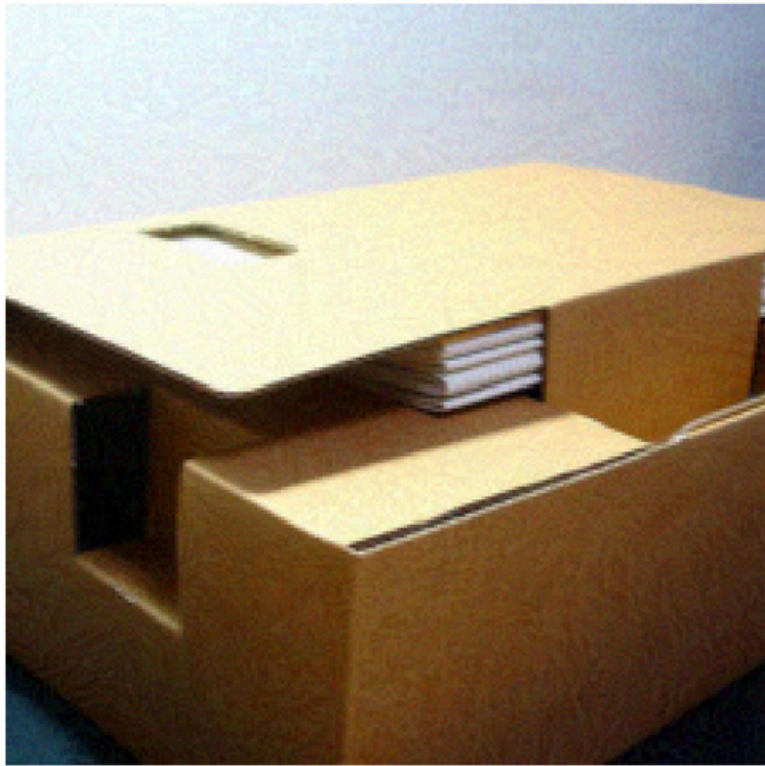
FGSM



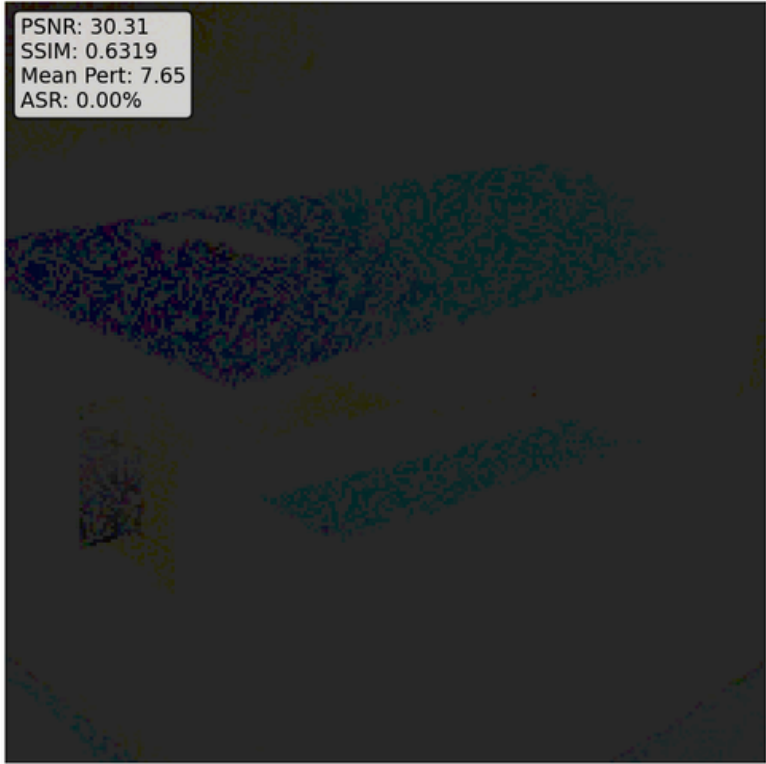
PGD

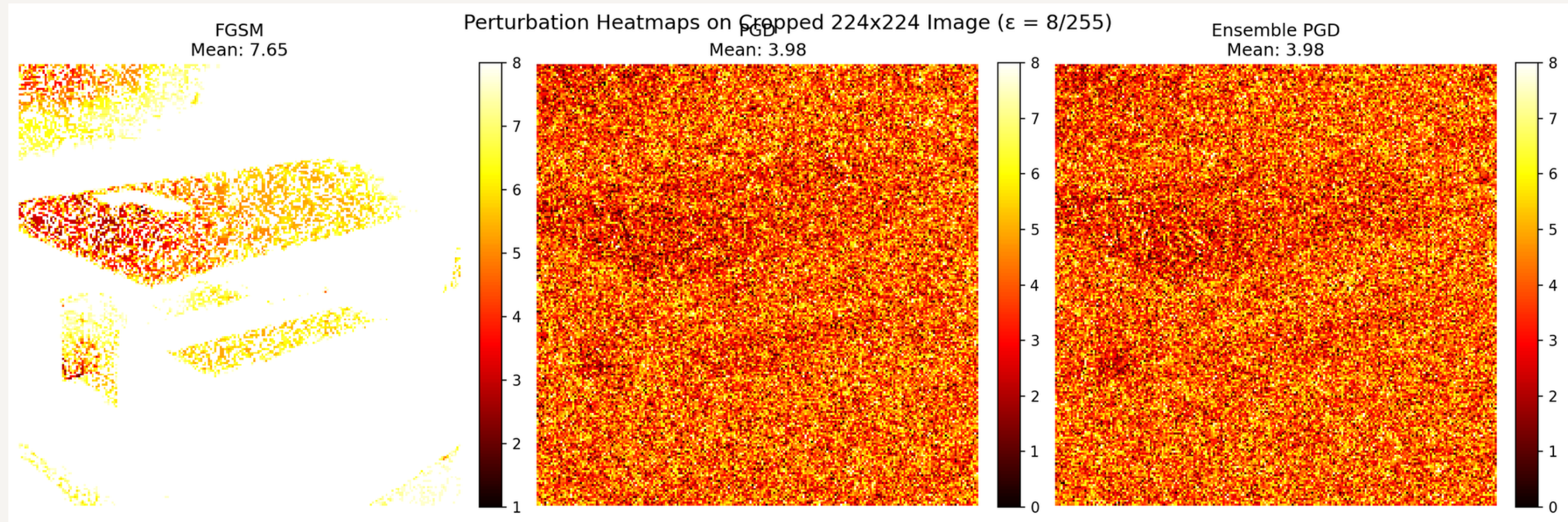


Ensemble PGD

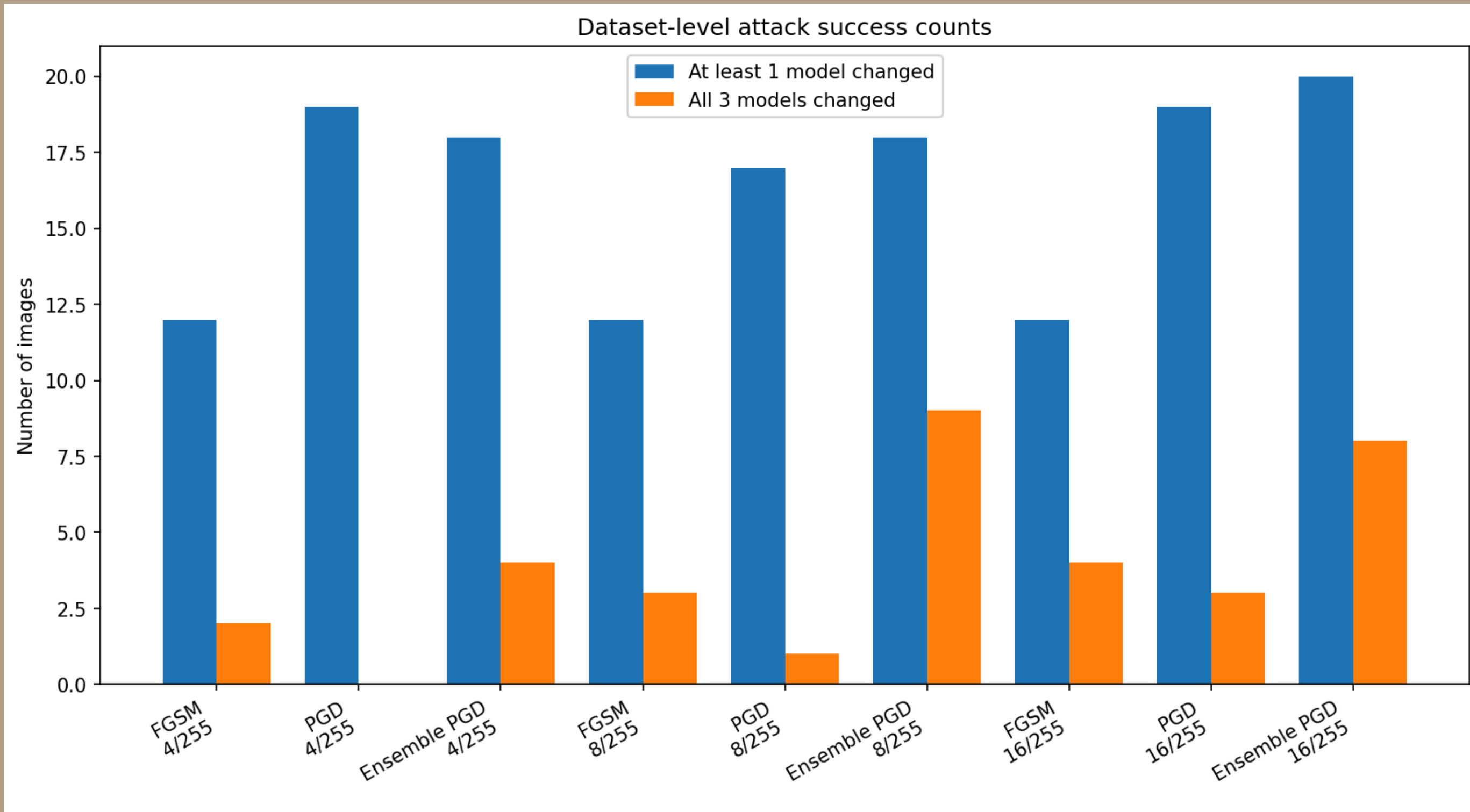


Perturbation Maps
(5x amplified)

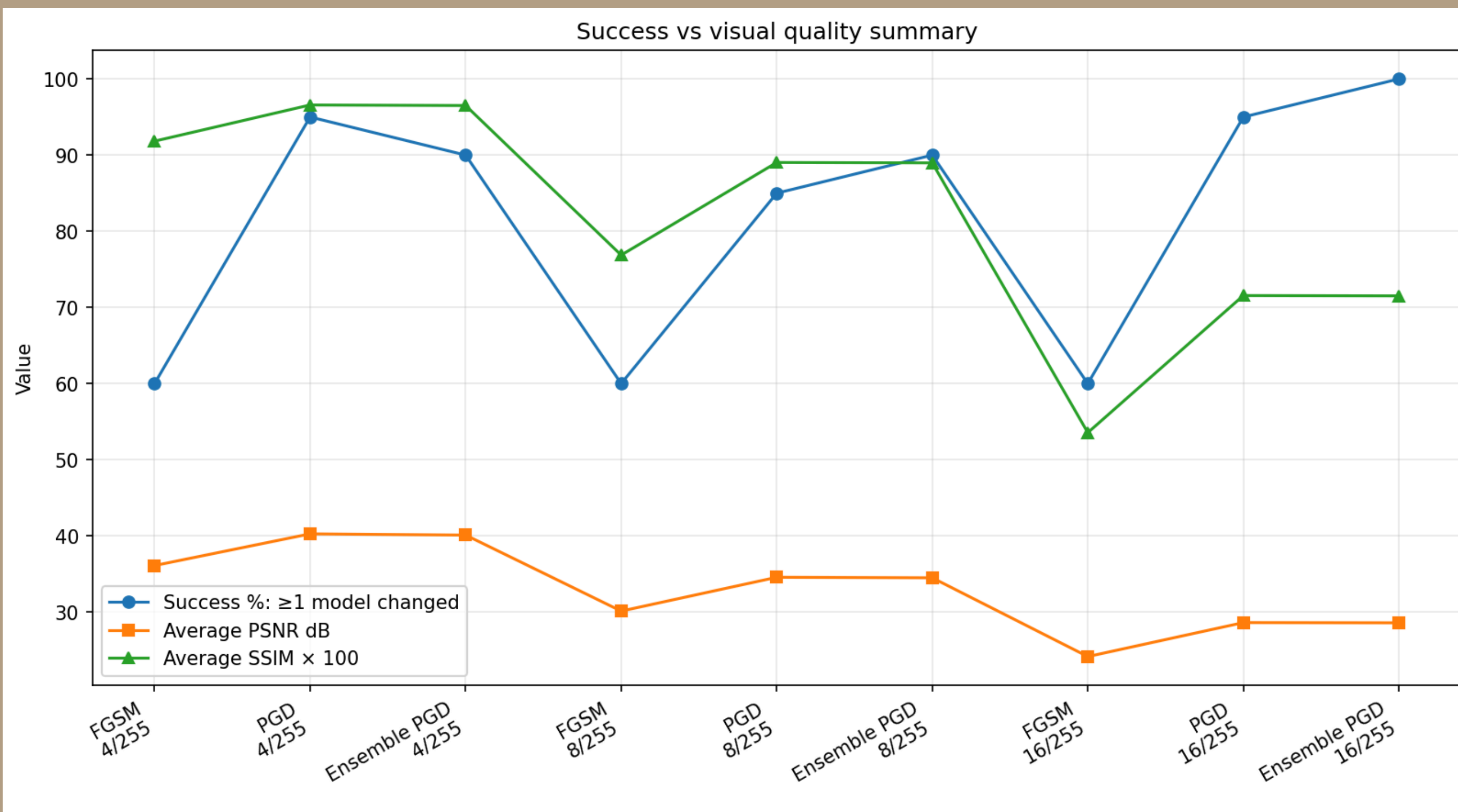




Experimental Setup Results



Experimental Setup Results



Experimental Setup II

- **Dataset:** 20 digital drawings
- **Epsilon values:** 4/255, 8/255, 16/255
- **PGD iterations:** 20
- Three evaluation models all classify drawings as comic books.

anonymous artist



Epsilon 8/255

Prediction Details FGSM

resnet50: comic book (33.46%) -> book jacket (11.73%) 

vit: comic book (80.65%) -> comic book (67.87%) 

efficientnet: comic book (81.48%) -> book jacket (41.57%) 

Prediction Details PGD

resnet50: comic book (33.46%) -> comic book (0.47%) 

vit: comic book (80.65%) -> comic book (80.40%) 

efficientnet: comic book (81.48%) -> comic book (45.22%) 

Prediction Details Ensemble PGD

resnet50: comic book (33.46%) -> lampshade (0.24%) 

vit: comic book (80.65%) -> trombone (20.38%) 

efficientnet: comic book (81.48%) -> perfume (3.61%) 

Attack Method Comparison on Cropped 224x224 Image ($\epsilon = 8/255$)

Original Cropped 224x224



FGSM



PSNR: 30.07
SSIM: 0.7398
Mean Pert: 8.00
ASR: 66.67%

PGD



PSNR: 34.54
SSIM: 0.8715
Mean Pert: 4.12
ASR: 0.00%

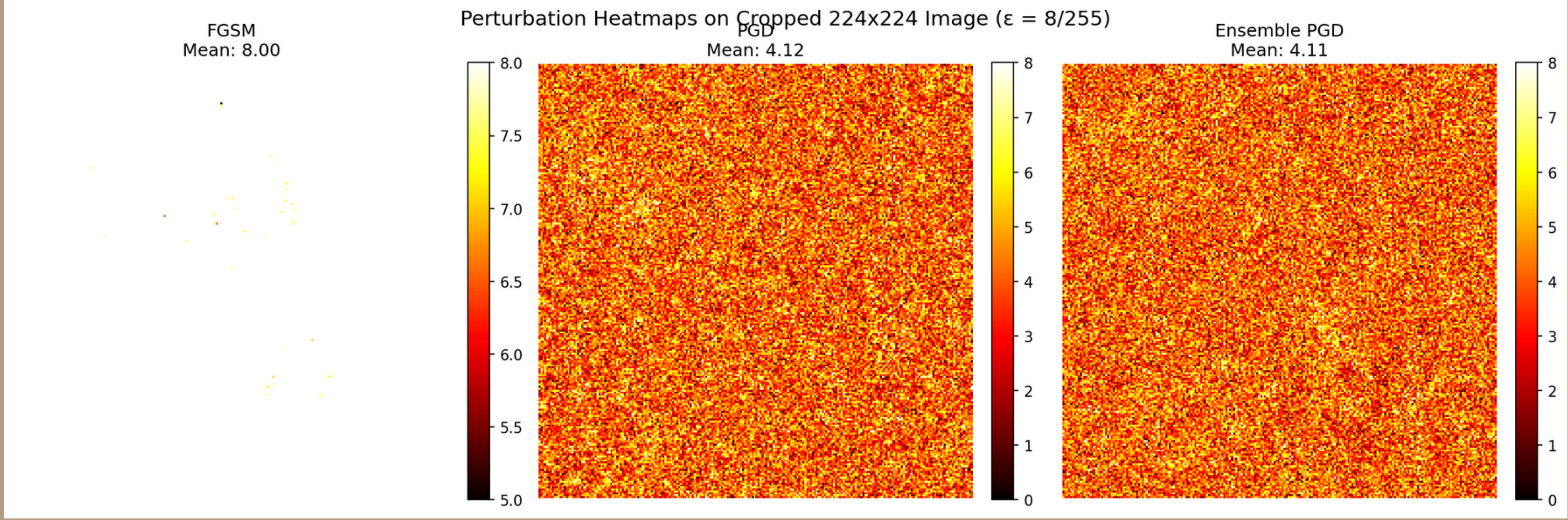
Ensemble PGD



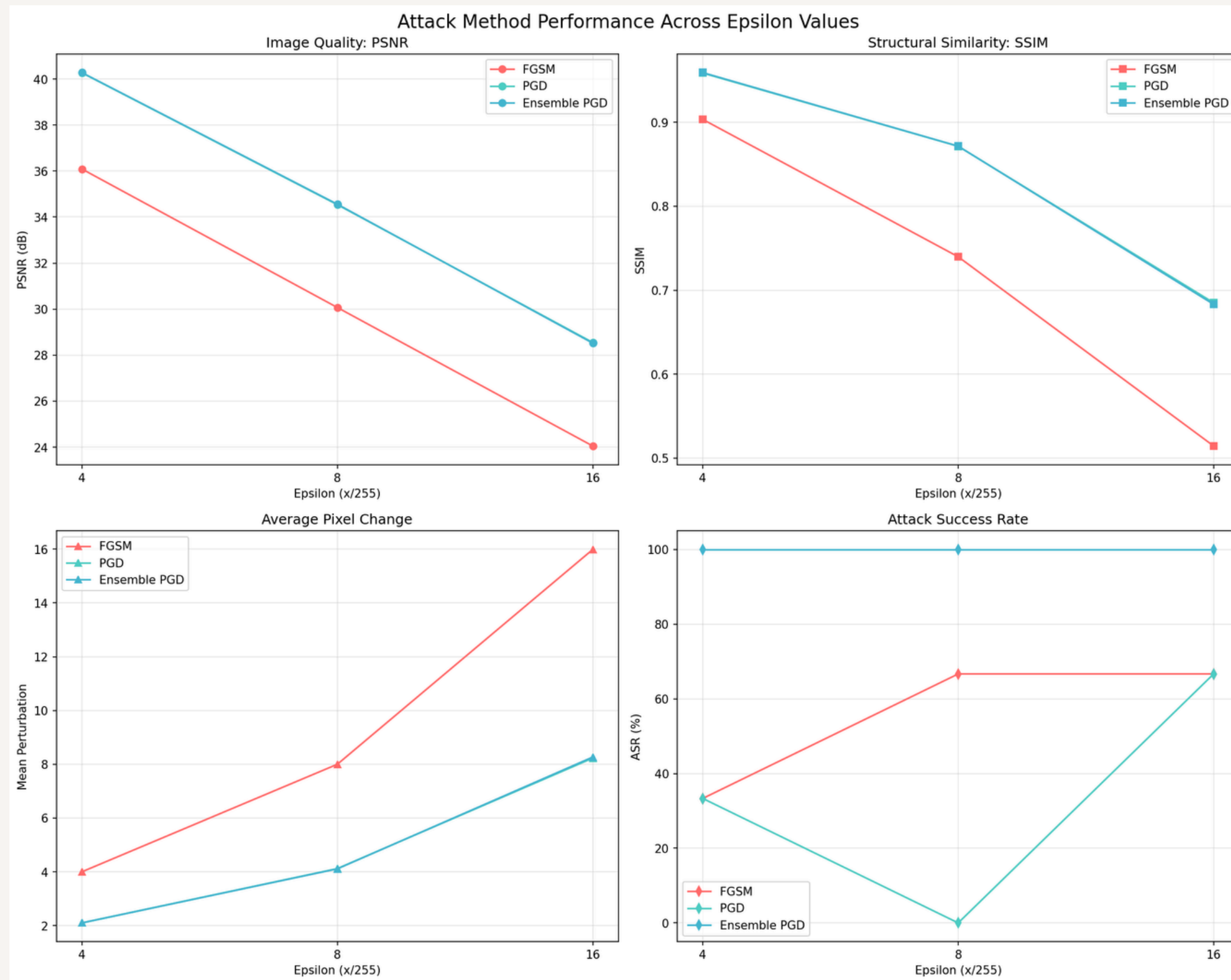
PSNR: 34.56
SSIM: 0.8716
Mean Pert: 4.11
ASR: 100.00%

Perturbation Maps
(5x amplified)

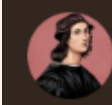




Experimental Setup || Results



Raphael AI Experiment



Raphael AI

original



Raphael AI

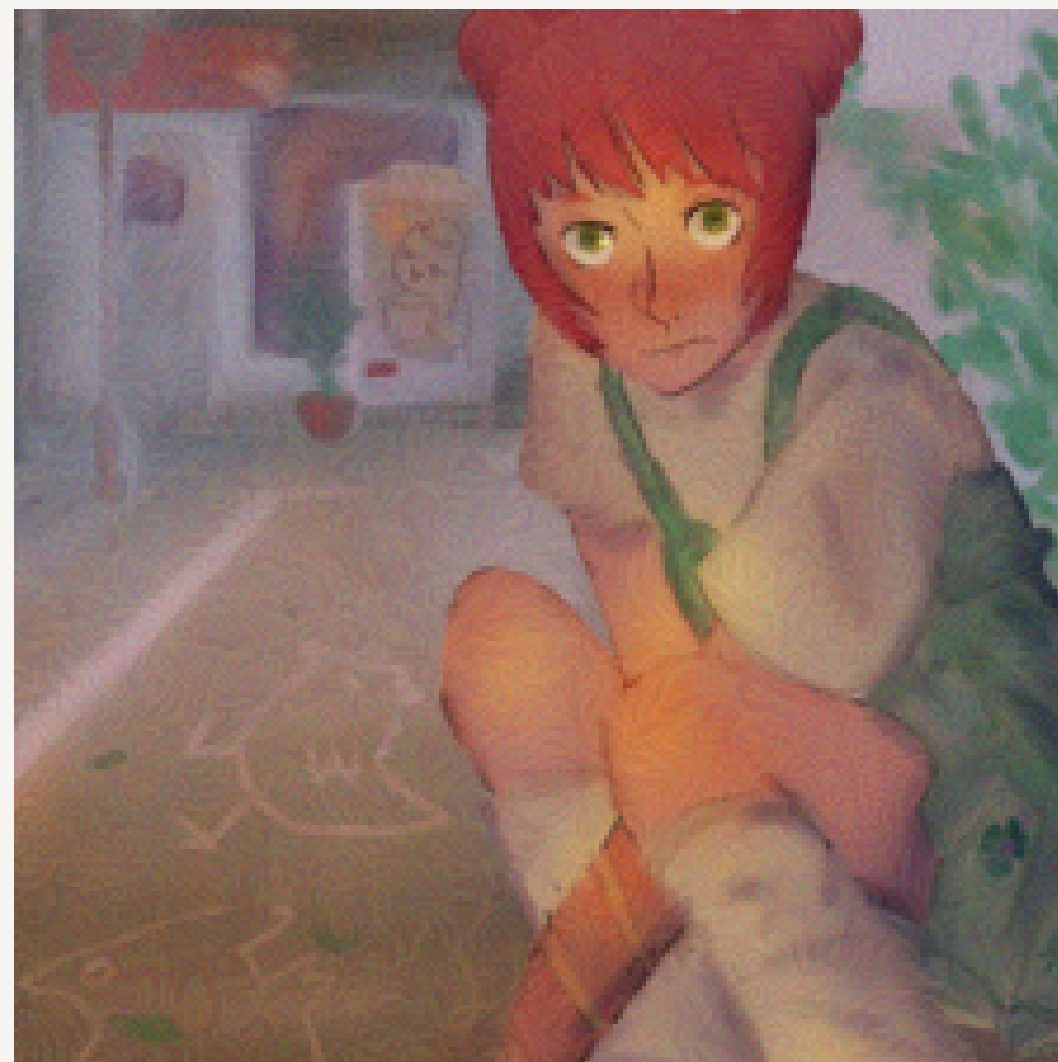


Epsilon 8/255

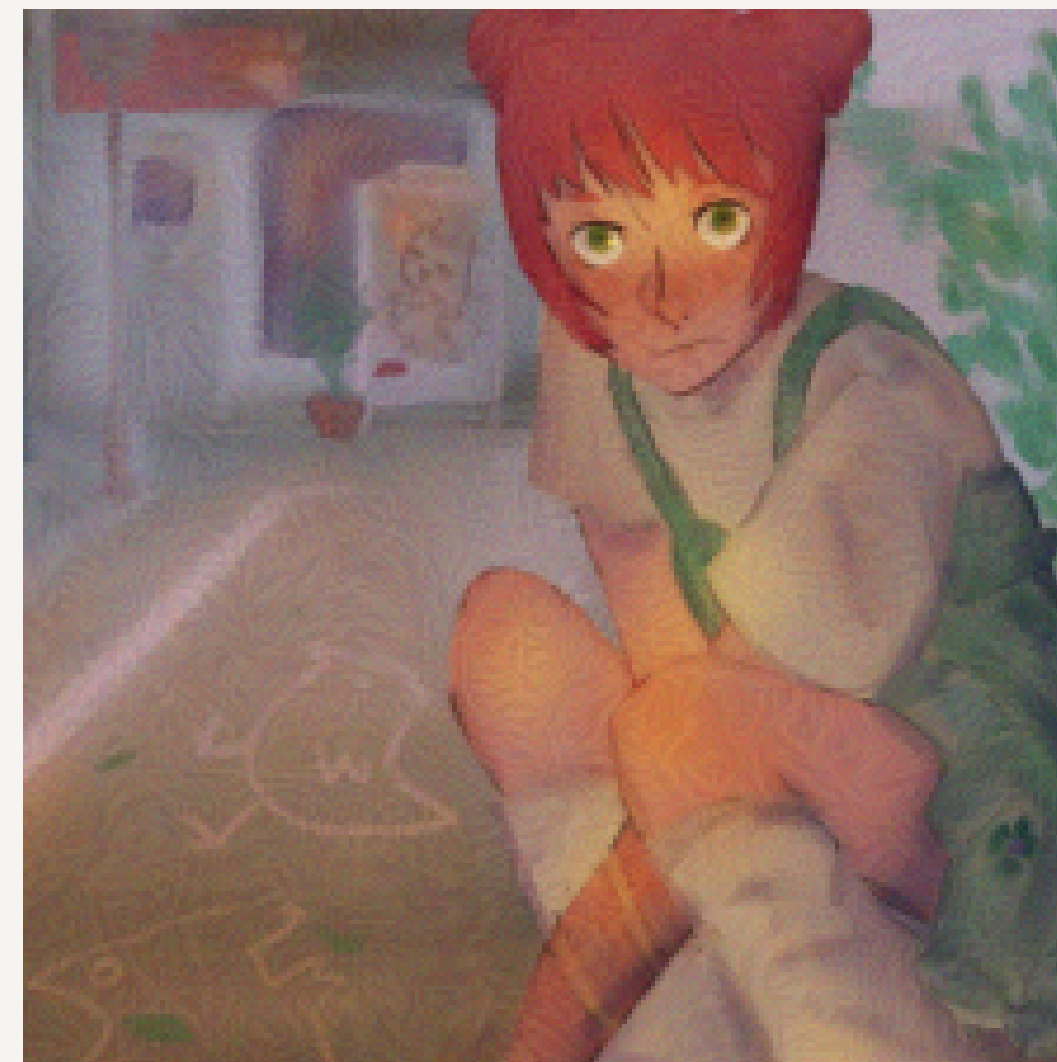
FGSM



PGD



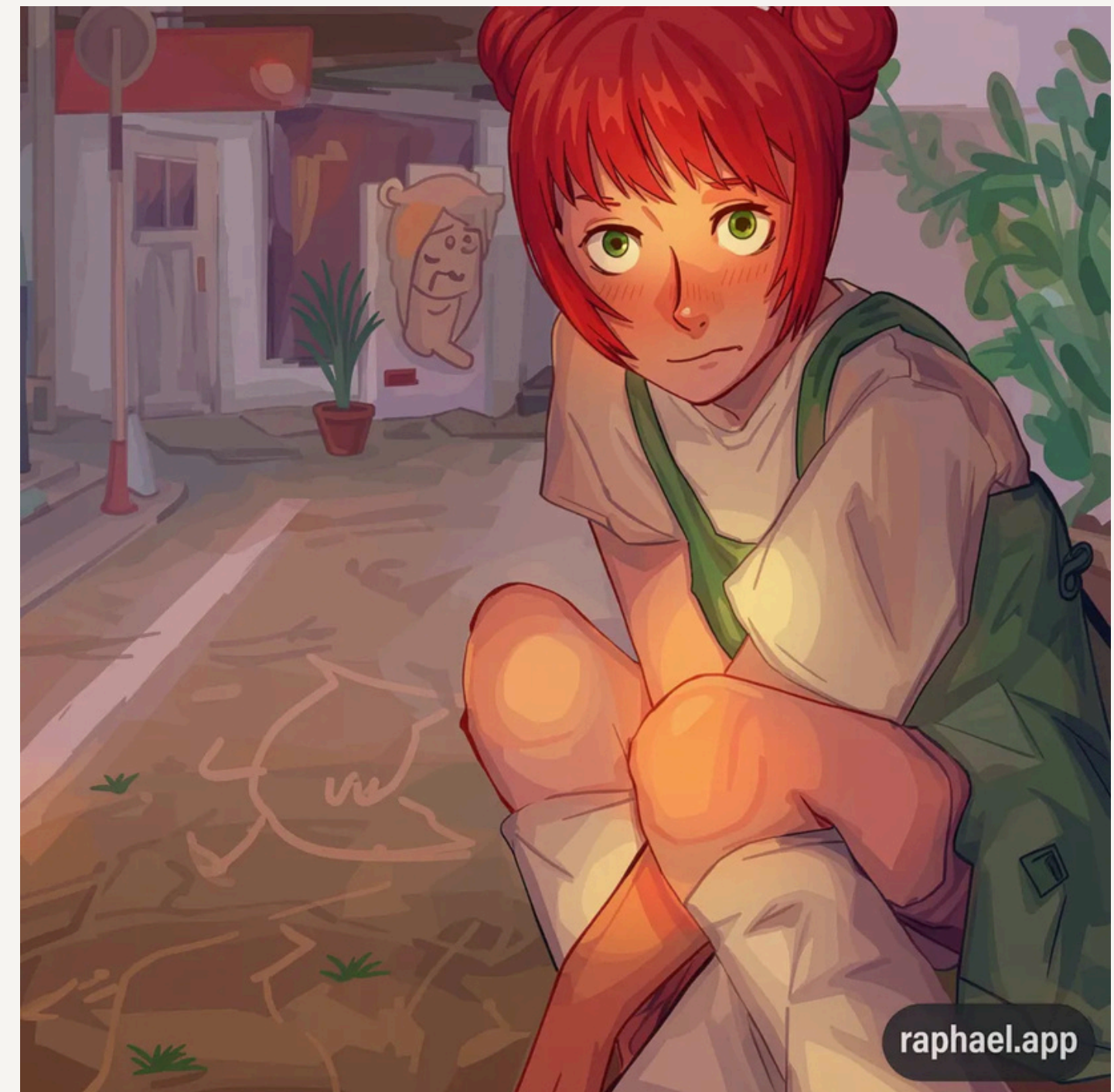
Ensemble
PGD



original



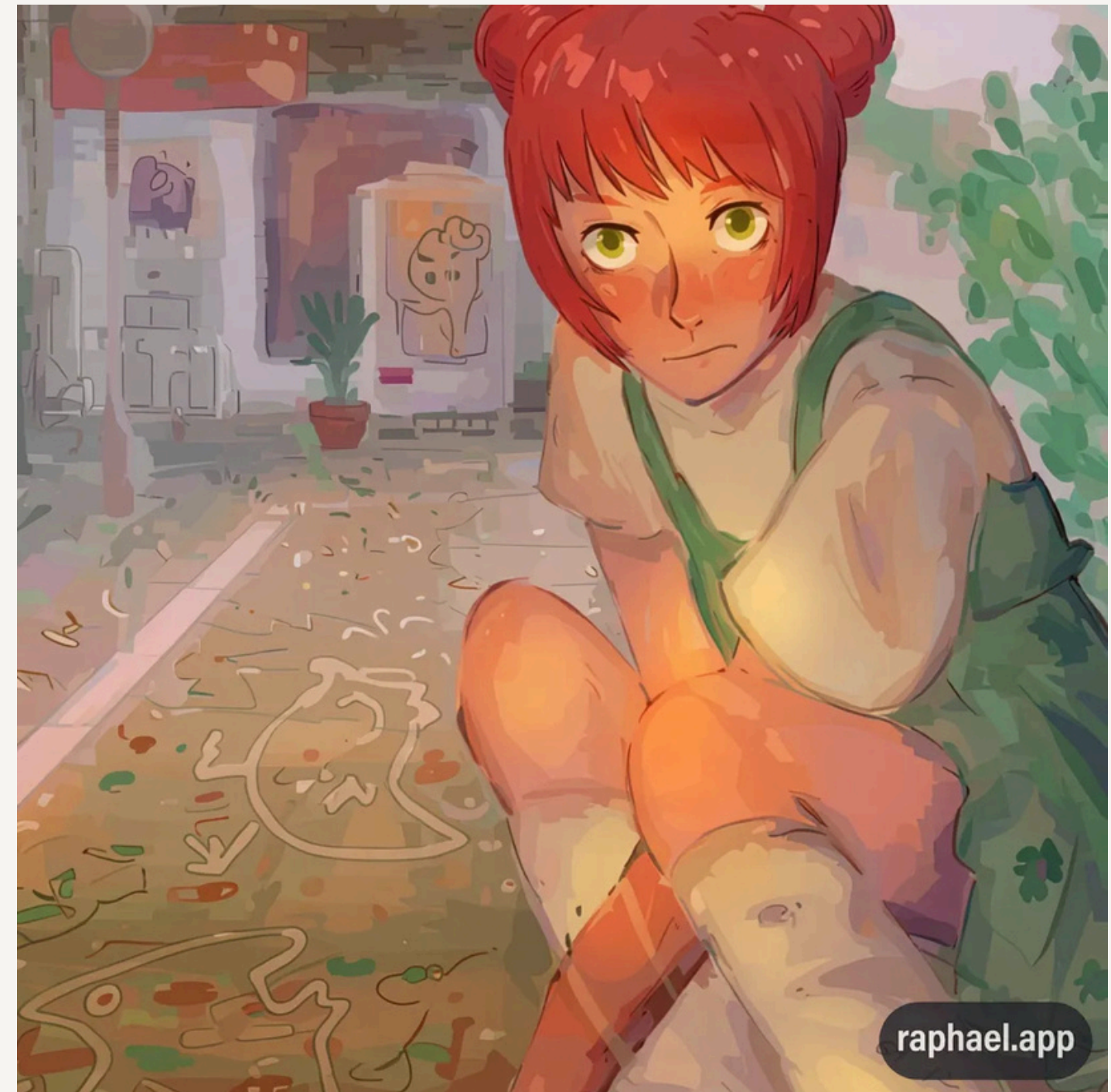
FGSM on
Raphael AI



original



PGD on
Raphael AI



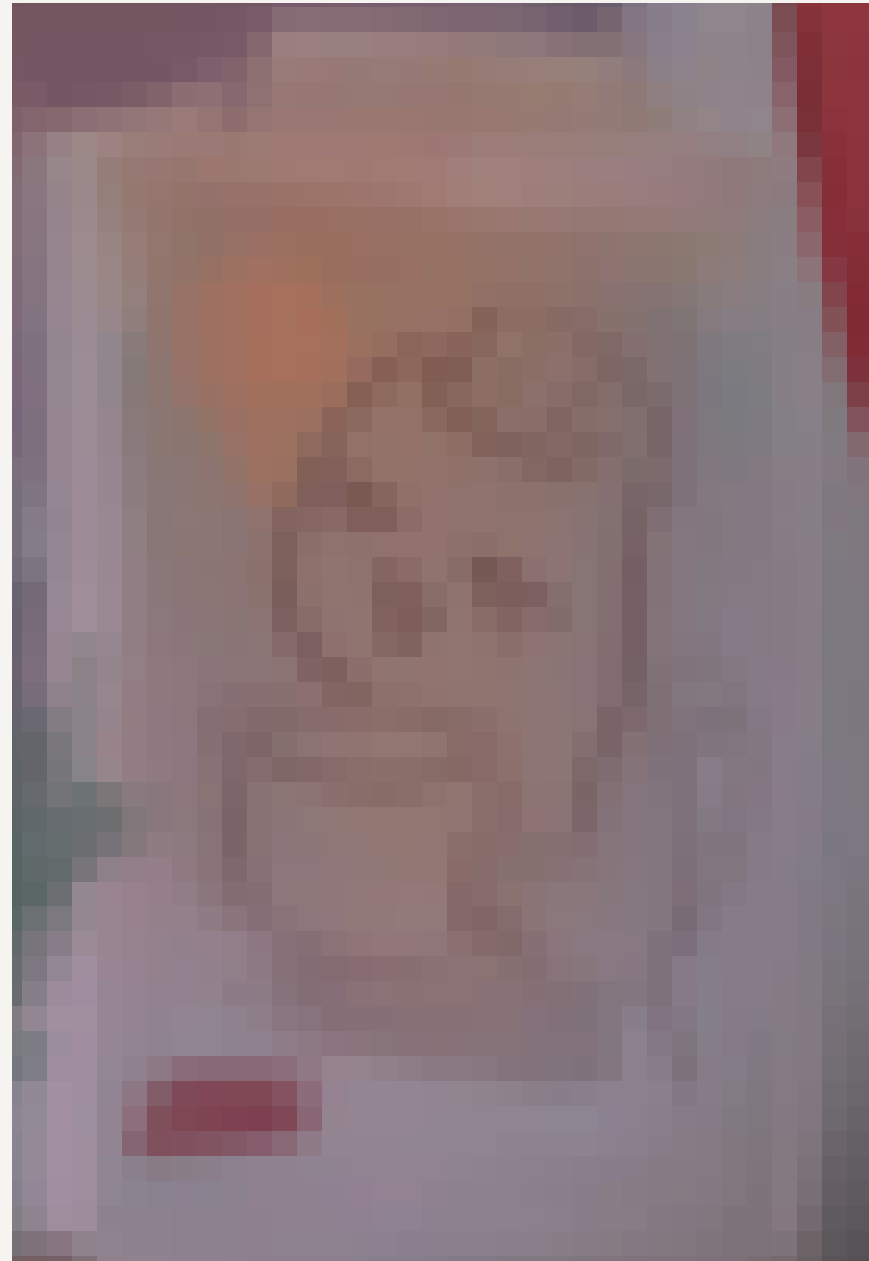
original



Ensemble PGD
on Raphael AI



original

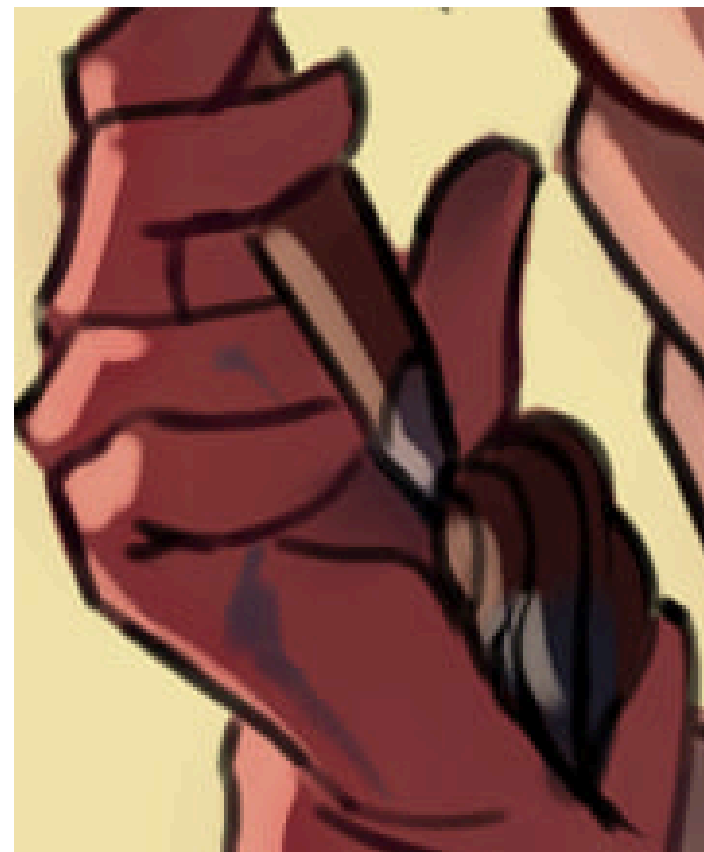


unprotected
on Raphael AI



Ensemble PGD
on Raphael AI





ORIGINAL IMAGE

1 blurred watercolor stain on pants

2 brush in hand,
light facing left sight.
Blurred purple spot on gloves

3 the ladybug has muddy colors



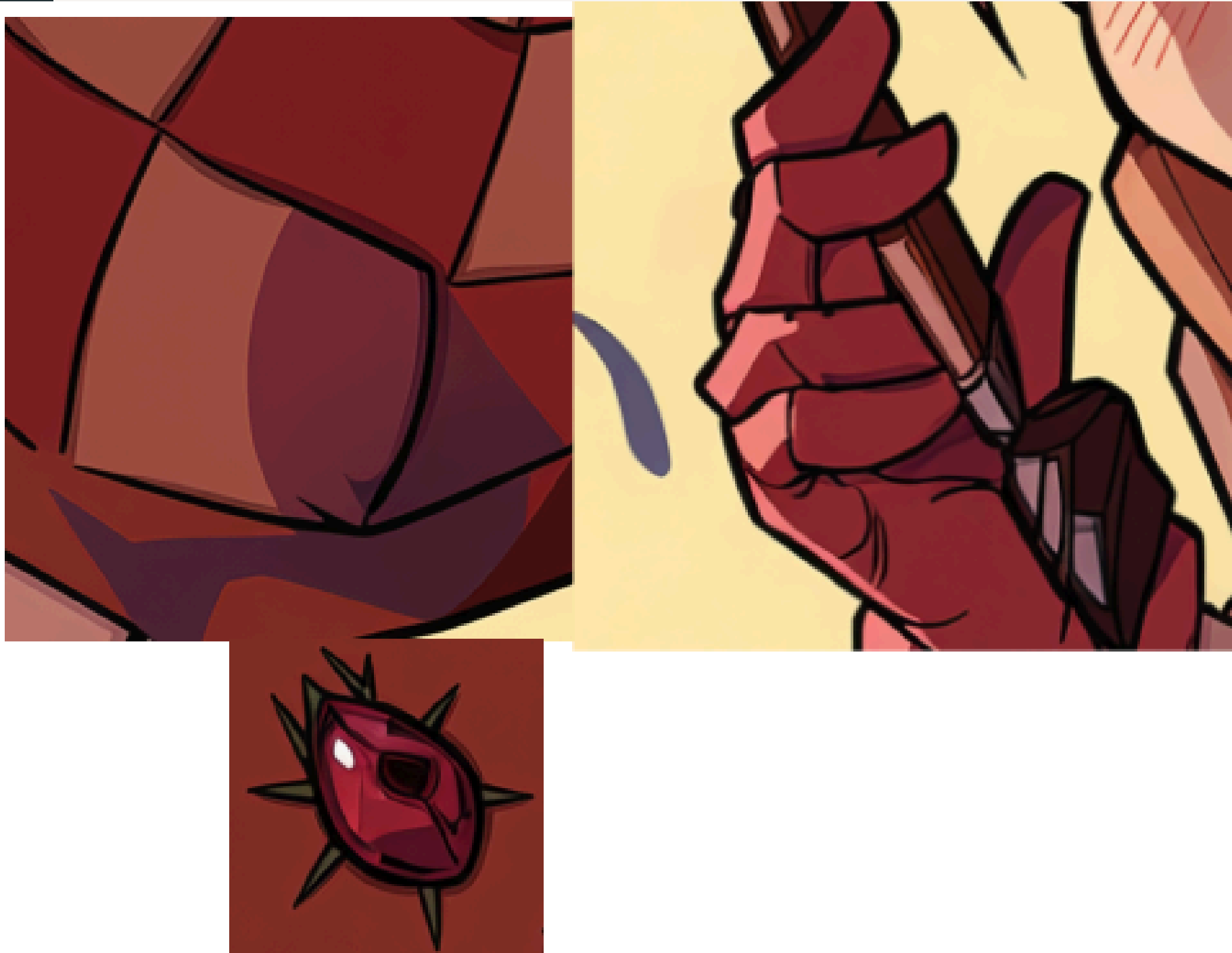


UNPROTECTED

1 stain is no longer blurr,
but has smudged textured edges

2 purple spot is there with
different shape

3
glare on a beetle
overall no visible changes



PGD EPSILON 16

1 stain is no longer blurr,
texture has been erased.

2 more lines apperead,
purple spot is no longer there
lineart is sharp

3 the ladybug has bright colors
it had changed the shape
glare on a beetle

- 01 FGSM provides fast but weaker adversarial attacks.
- 02 PGD improves attack performance through iterative optimization.
- 03 Ensemble PGD achieves the highest cross-model transferability.
- 04 Combining CNN and Transformer architectures produces stronger transferable perturbations.
- 05 Additional experiments suggest that these perturbations alone are insufficient to reliably protect against diffusion-based generative AI.

Conclusion



References

- [1] Ren et al., “Adversarial Attacks and Defenses in Deep Learning,” Engineering, 2020.
- [2] Shan et al., “Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models,” USENIX Security, 2023.
- [3] Zhao et al., “Data Poisoning in Deep Learning: A Survey,” arXiv, 2025.
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- [5] Wang et al., “A Survey on Large Language Model Based Autonomous Agents,” arXiv, 2025.
- [6] Sohn et al., “Visual Prompt Tuning for Generative Transfer Learning,” ECCV, 2022.
- [7] Hönig et al., “Adversarial Perturbations Cannot Reliably Protect Artists From Generative AI,” arXiv, 2024.



Thank You

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